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Lithium ion conduction in A-site deficient perovskites $R_{1/4}Li_{1/4}TaO_3$ (R = La, Nd, Sm and Y)

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Abstract

New lithium ion conductors of perovskite-type compounds $R_{1/4}Li_{1/4}TaO_3$ (R = La, Nd, Sm and Y) were synthesized, and their crystal structure and lithium ion conductivity were characterized. Symmetry of the perovskite lattice changed from cubic for $La_{1/4}Li_{1/4}TaO_3$ to tetragonal for $Nd_{1/4}Li_{1/4}TaO_3$ and $Sm_{1/4}Li_{1/4}TaO_3$, and to orthorhombic for $Y_{1/4}Li_{1/4}TaO_3$. The alternate ordering of R and Li ions and vacancies at the A-sites was observed for $Nd_{1/4}Li_{1/4}TaO_3$, $Sm_{1/4}Li_{1/4}TaO_3$ and $Y_{1/4}Li_{1/4}TaO_3$. The cube root ($V^{1/3}$) of the subcell volume decreased with decreasing the ionic radius of R cations which act as spacers for the perovskite lattice. With decrease in $V^{1/3}$ from $La_{1/4}Li_{1/4}TaO_3$ to $Y_{1/4}Li_{1/4}TaO_3$, the bulk conductivity at 400 K decreased from 1.4×10^{-1} S m^{-1} to 5.0×10^{-7} S m^{-1} and its activation energy increased from 0.35 eV to 0.85 eV. The subcell volume and the ordering at the A-sites are major factors in influencing lithium ion conduction in these perovskite-type conductors. © 1999 Published by Elsevier Science B.V. All rights reserved.

Keywords: Lithium ion conductor; Perovskite; Lanthanide ions; Lithium ion mobility; Activation energy

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1. Introduction

The lithium ion conductor is a key material for solid electrolytes in high energy density batteries and electrochemical devices. Recently, the lithium ion conductors with perovskite-related structures such as $La_{2/3-x}Li_{3x}TiO_3$ [1–9], $Ln_{1/2}Li_{1/2}TiO_3$ (Ln = La, Pr, Nd and Sm) [3,4,8], $La_{1/3-x}Li_{3x}NbO_3$ [6] and $Li_{2x}Sr_{1/2x}M_{1/2-x}Ta_{1/2+x}O_3$ (M = Cr, Fe, Co, Ga, In) [16] have been investigated. The presence of many equivalent A-sites available for lithium ion migration leads to their high lithium ion conductivity. The perovskite-type compounds containing Ti^{4+} ions at the B-sites are easily reduced and become electronic conductors when they contact with lithium metal, although they show a high lithium ion conductivity at room temperature [1]. In contrast, those containing Ta^{5+} ions at the B-sites are less susceptible to reduction [16]. Furthermore, the A-site deficient tantalate $La_{1/3-x}Li_{3x}TaO_3$ inherently possesses much more vacancies at the A-sites than the titanate $La_{2/3-x}Li_{3x}TiO_3$.

We have prepared the perovskite-type solid solutions $La_{1/3-x}Li_{3x}TaO_3$ ($x \le 1/6$) derived from $La_{1/3}TaO_3$ and characterized the crystal structure,

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lithium ion conductivity and activation energy for lithium ion conduction. The maximum bulk conductivity of 7×10^{-3} S m^{-1} was observed at room temperature for the sample with a composition of $x = 0.06$ [10]. Moreover, lithium ion mobility and activation energy in the perovskites $La_{1/3-x}Li_{3x}TaO_3$ have been found to depend on the cube root of the subcell volume and the degree of alternate ordering of cations and vacancies at the A-sites [11].

In the present paper, we report the crystal structure and the lithium ion conductivity of the A-site deficient perovskites $R_{1/4}Li_{1/4}TaO_3$ where R is La, Nd, Sm and Y.

2. Experimental

La_2O_3(4N), Nd_2O_3(3N), Sm_2O_3(3N), Li_2CO_3(3N), Ta_2O_5(3N) (Nacalai Tesque) and Y_2O_3(4N) (Kishida Chemical Co.) were used as starting materials. Reagents of La_2O_3, Nd_2O_3, Sm_2O_3 and Y_2O_3 were pre-fired at 773 K for 6 h before use and stored in a desiccator. A stoichiometric mixture of starting materials was calcined in air at 973 K for 12 h and then ground. The powder was uniaxially pressed at 30 MPa and then isostatically pressed at 200 MPa into a pellet 15 mm in diameter and 2 mm in thickness. The pellet was put into a Pt container together with the calcined powder of the same composition and fired twice at 1773 K for 48 h in a covered alumina crucible with intermediate grinding. Sintered samples were polished and then characterized as follows.

Identification of crystalline phases and determination of the lattice parameter were carried out by X-ray powder diffraction (XRD) analysis using a Rigaku RAD-rA diffractometer with CoK_α radiation and by the CELL-series programs [19].

Chemical analysis was carried out by inductively coupled plasma (ICP) spectroscopy (SPS 1200A, Seiko Instruments) and atomic absorption spectroscopy (AA-630-01, Shimadzu), in order to determine metal contents in the sample.

For the electrical conductivity measurement, Au paste was painted on both faces of sample and baked as blocking electrodes. The AC impedance of the sample was measured by using a YHP4192A impedance analyzer in air over the temperature range from room temperature to 673 K and the frequency range from 5 Hz to 13 MHz. The collected data were analyzed by the complex impedance plane method in order to determine the bulk conductivity of the sample.

3. Results and discussion

3.1. Crystal structure

Single phase solid solutions were prepared by solid state reaction under the above conditions, and their compositions were found to be $La_{0.25}Li_{0.23}TaO_{2.99}$, $Nd_{0.24}Li_{0.24}TaO_{2.99}$, $Sm_{0.23}Li_{0.24}TaO_{2.97}$ and $Y_{0.24}Li_{0.23}TaO_{2.98}$, respectively, according to chemical analysis. The oxygen content of the sample was calculated from the electroneutrality. These results indicate that the loss of Li_2O during firing was below 8%.

XRD data are given in Tables 1–4 for the perovskites $La_{1/4}Li_{1/4}TaO_3$, $Nd_{1/4}Li_{1/4}TaO_3$, $Sm_{1/4}Li_{1/4}TaO_3$ and $Y_{1/4}Li_{1/4}TaO_3$, respectively. Here, these four perovskites are abbreviated, respectively, as LaLT, NdLT, SmLT and YLT, and $R_{1/4}Li_{1/4}TaO_3$ is as RLT. In the XRD patterns of NdLT, SmLT and YLT, weak superstructure lines were observed and they are asterisked in Tables 1–4. Fig. 1 shows details of XRD patterns at $2\theta = 92–96°$ for RLT. With decreasing the ionic radius of R, line splitting was observed, indicating the reduction of

Table 1
XRD data for the perovskite $La_{1/4}Li_{1/4}TaO_3$

hkl	d_{calc}/pm	d_{obs}/pm	I_{obs}
100	390.6	390.4	92
110	276.2	276.4	100
111	225.5	225.7	12
200	195.3	195.4	34
210	174.7	174.7	38
211	159.5	159.5	32
220	138.1	138.1	14
221, 300	130.2	130.2	19
310	123.5	123.5	14
311	117.8	117.8	8
222	112.8	112.8	7
320	108.3	108.3	7
321	104.4	104.4	23

Pm3m, cubic, $a = 390.63$ pm.

Table 2
XRD data for the perovskite $Nd_{1/4}Li_{1/4}TaO_3$

hkl	d_{calc}/pm	d_{obs}/pm	I_{obs}
001*	773.0	777.8	3
100	387.2	387.2	88
002	386.5		
101*	346.2	346.4	6
110	273.8	273.7	100
102	273.5		
111*	258.1	259.0	1
003*	257.7		
112	223.4	223.5	12
103*	214.5	–	–
200	193.6	193.5	34
004	193.3		
201*	187.8	187.8	2
113*	187.6		
210	173.2	173.1	35
202	173.1		
104	172.9		
211*	169.0	168.8	1
212	158.0	158.0	44
114	157.9		
203*	154.8	155.0	1
005*	154.6		
213*	143.7	143.9	1
105*	143.6		
220	136.9	136.8	14
204	136.8		
221*	134.8	–	–
115*	134.6	–	–
300	129.1	129.0	16
222,214	129.0		
006	128.8		
301*	127.3	–	–
310, 302	122.4	122.7	7
106	122.2	122.4	14

P4/mmm, tetragonal, $a = 387.20$ pm, $c = 773.00$ pm.
*Superstructure lines; –, not observed.

Table 3
XRD data for the perovskite $Sm_{1/4}Li_{1/4}TaO_3$

hkl	d_{calc}/pm	d_{obs}/pm	I_{obs}
001*	771.0	775.3	2
100	385.9	385.9	81
002	385.5		
101*	345.1	345.2	6
110	272.9	272.7	100
102	272.7		
111*	257.2	257.8	2
003*	257.0		
112	222.7	222.8	11
103*	213.9	–	–
200	193.0	192.8	33
004	192.8		
201*	187.2	187.2	4
113*	187.1		
210	172.6	172.4	32
202	172.5		
104	172.4		
211*	168.4	168.2	1
212	157.5	157.4	39
114	157.4		
203*	154.3	154.3	1
005*	154.2		
213*	143.3	143.3	2
105*	143.2		
220, 204	136.4	136.3	13
221*	134.4	–	–
115*	134.3	–	–
300,222,214	128.6	128.5	13
006	128.5		
301*	126.9	–	–
310, 302	122.0	122.2	8
106	121.9	121.9	15
311, 233, 205*	120.5	–	–
312, 116	116.3	116.2	6

P4/mmm, tetragonal, $a = 385.90$ pm, $c = 771.00$ pm.
*Superstructure lines; –, not observed.

symmetry: the splittings into two and three peaks correspond to the tetragonal and orthorhombic structures, respectively. These superstructure lines are ascribed to the alternate ordering of R and Li ions and vacancies at the A-sites along the *c*-axis [4,10,13,20]. However, another type of superstructure lines due to the ordering of R and Li ions and vacancies with the rocksalt arrangement reported for $Ln_{1/2}Li_{1/2}TiO_3$ [4] was not observed. This indicates that the distribution of R and Li ions and vacancies in each A-site layer normal to the *c*-axis is random for the perovskites RLT. The symmetry of the perovskite lattice changed from cubic for LaLT to tetragonal for NdLT and SmLT, and to orthorhombic for YLT. Possible space groups are considered to be Pm3m for LaLT, P4/mmm for NdLT and SmLT, and Pmmm for YLT.

No perovskite-type compounds containing lithium

Table 4
XRD data for the perovskite $Y_{1/4}Li_{1/4}TaO_3$

hkl	d_{calc}/pm	d_{obs}/pm	I_{obs}
001*	765.4	765.2	2
010	384.2	384.5	51
002	382.7	382.7	79
100	382.2		
011*	343.4	342.2	8
101*	341.9		
012	271.2	270.6	100
110	271.0		
102	270.4		
111*	255.4	255.9	1
003*	255.1		
112	221.2	220.9	17
013*	212.5	–	–
103*	212.2	–	–
020	192.1	192.1	14
004	191.4	191.1	35
200	191.1		
021*	186.3	185.9	3
113*	185.8		
201*	185.4		
022, 120	171.7	171.7	24
014	171.3	171.0	42
104,210	171.1		
202	171.0		
121*	167.5	166.8	1
211*	167.0		
122	156.6	156.6	22
114	156.3	156.1	46
212	156.2		
023*	153.5	153.1	1
005*	153.1		
203*	153.0		

Pmmm, orthorhombic.
$a = 382.20$ pm, $b = 384.22$ pm, $c = 765.40$ pm.
*Superstructure lines; –, not observed.

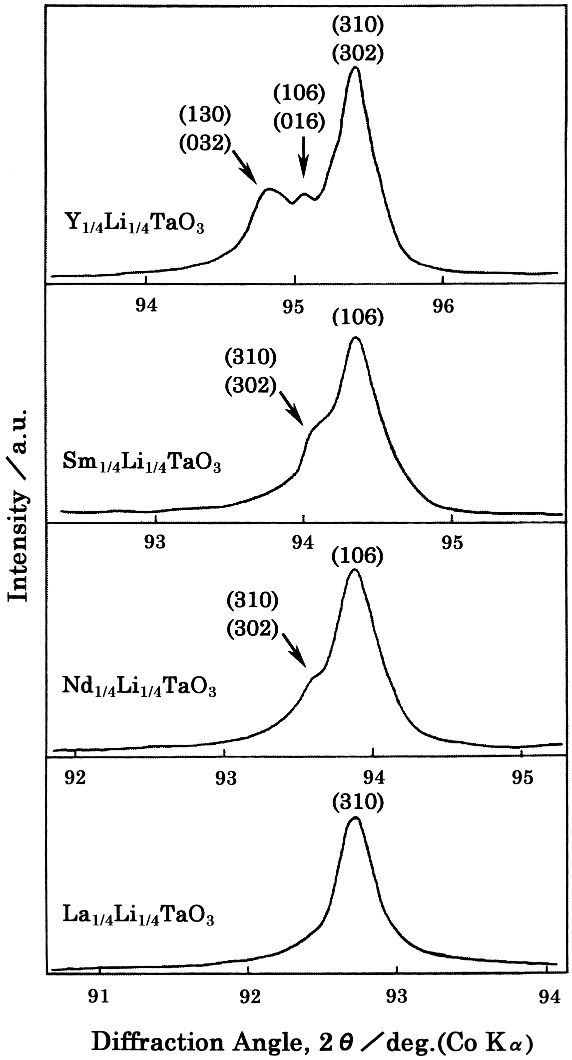

Fig. 1. XRD patterns at $2\theta = 92–96°$ for the perovskites $R_{1/4}Li_{1/4}TaO_3$.

ions alone at the A-sites have been known. On the assumption that lithium ions do not act as spacers in the framework of the perovskite lattice, tolerance factors for LaLT, NdLT, SmLT and YLT are evaluated to be 0.957, 0.925, 0.915 and 0.898, respectively, from reported ionic radii of La^{3+}, Nd^{3+}, Sm^{3+}, Ta^{5+}, O^{2-} and estimated ionic radius (119 pm) of Y^{3+} in 12-coordination by extrapolating the ionic radius vs. coordination number plots [12]. The tolerance factor values for NdLT, SmLT and YLT rationalize their tetragonal and orthorhombic distortion. Lowering of the symmetry and the alternate ordering of R and Li ions and vacancies at the A-sites presumably cause distortion of the bottleneck for lithium ion transport.

Fig. 2 shows the relationship between the cube

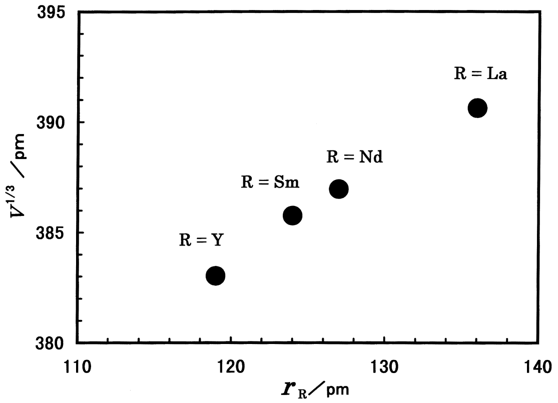

Fig. 2. Relationship between the cube root ($V^{1/3}$) of the subcell volume and ionic radius (r_R) of R ions for the perovskites $R_{1/4}Li_{1/4}TaO_3$.

root ($V^{1/3}$) of the subcell volume and the ionic radius (r_R) of lanthanide and yttrium ions in 12-coordination [12]. $V^{1/3}$ of RLT decreases almost linearly with decreasing the ionic radius of R, like in the perovskite-type rare-earth and yttrium tantalates ($R_{1/3}TaO_3$, R = La, Ce, Pr, Nd, Sm, Gd, Dy, Y) [13,20]. We have already reported that $V^{1/3}$ decreases with increasing x in the perovskites $La_{1/3-x}Li_{3x}TaO_3$ ($x \leq 1/6$) [10,11]. Thus, the subcell volume of the tantalate-based perovskite oxides can be controlled by the size and the content of spacer cations.

3.2. Ionic conductivity and activation energy for lithium ion conduction

Fig. 3 shows the temperature dependence of the bulk conductivity for RLT in the Arrhenius formalism. Deviation from the linear relation is observed in the high temperature range of LaLT. Such non-Arrhenius behavior has been reported for crystalline fast ion conductors [17,18] and glassy fast ion conductors [14,15]. This non-Arrhenius behavior is considered to be a general feature of fast ion conductors and to be a result of the correlation effects between mobile ions. The study on dynamics of ion transport in $La_{1/3-x}Li_{3x}TaO_3$ and RLT is now in progress.

Fig. 4 shows the lithium ion conductivity at 400 K and the activation energy for lithium ion conduction as a function of $V^{1/3}$. With increase in $V^{1/3}$ from

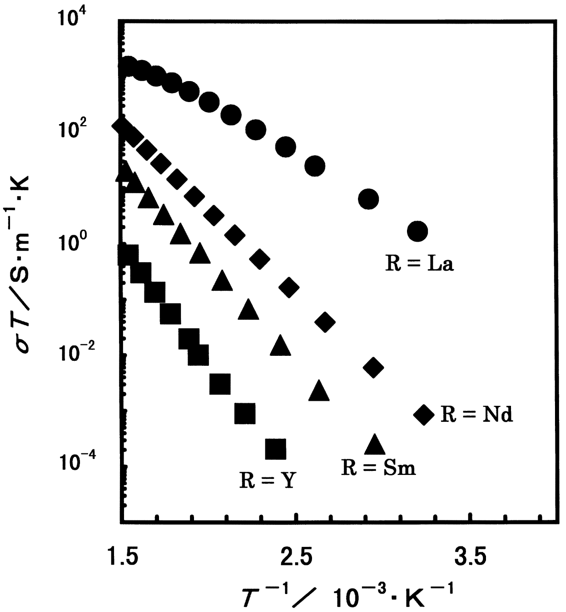

Fig. 3. Arrhenius plots of the bulk conductivity for the perovskites $R_{1/4}Li_{1/4}TaO_3$.

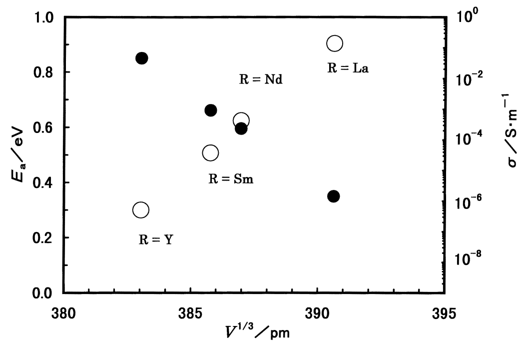

Fig. 4. Activation energy, E_a (●), for lithium ion conduction and lithium ion conductivity, σ (○), at 400 K as a function of the cube root of the subcell volume, $V^{1/3}$, for the perovskites $R_{1/4}Li_{1/4}TaO_3$.

YLT to LaLT, the activation energy decreases from 0.85 eV to 0.35 eV and the lithium ion conductivity increases from 5.0×10^{-7} S m^{-1} to 1.4×10^{-1} S m^{-1}. A similar phenomenon has been observed in the perovskites $La_{1/3-x}Li_{3x}TaO_3$ ($x \leq 1/6$) [11].

The results obtained for the perovskites $R_{1/4}Li_{1/4}TaO_3$ and $La_{1/3-x}Li_{3x}TaO_3$ ($x \leq 1/6$) [11] are summarized in Fig. 5. The mobility was calculated from Eq. (4) in Ref. [11]. The solid and broken

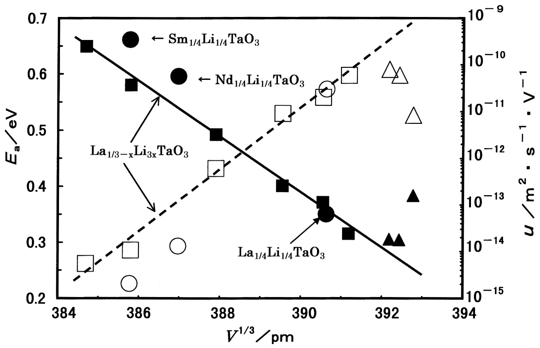

Fig. 5. Activation energy, E_a (●, ■, ▲), for lithium ion conduction and lithium ion mobility, u (○, □, △), at room temperature as a function of the cube root of the subcell volume, $V^{1/3}$, for the perovskite-type lithium ion conductors. ●, ○: $R_{1/4}Li_{1/4}TaO_3$ (R=La, Nd and Sm); ■, □: cubic $La_{1/3-x}Li_{3x}TaO_3$; ▲, △: tetragonal $La_{1/3-x}Li_{3x}TaO_3$.

lines in Fig. 5 indicate variations in the activation energy and lithium ion mobility, respectively, which are determined from data on cubic $La_{1/3-x}Li_{3x}TaO_3$ [11]. It can be said that the decrease in $V^{1/3}$ causes the contraction of the bottle-neck for lithium ion transport in perovskite-type ion conductors. Furthermore, a tendency to deviate from the linear relations becomes remarkable with increasing the ordering of R and Li ions and vacancies at the A-sites and the lattice distortion. Thus, the subcell volume and the ordering at the A-sites are the predominant factors which affect the conduction process in RLT and $La_{1/3-x}Li_{3x}TaO_3$.

4. Conclusions

We prepared new perovskites $La_{1/4}Li_{1/4}TaO_3$, $Nd_{1/4}Li_{1/4}TaO_3$, $Sm_{1/4}Li_{1/4}TaO_3$ and $Y_{1/4}Li_{1/4}TaO_3$ by solid state reaction and studied the effects of the subcell volume on the lithium ion conductivity and the activation energy for lithium ion conduction; the conclusions are as follows.

1. The substitution of smaller Nd^{3+}, Sm^{3+}, or Y^{3+} for La^{3+} in $La_{1/4}Li_{1/4}TaO_3$ decreased the subcell volume, and led to the alternate ordering of cations and vacancies at the A-sites. The symmetry was orthorhombic for $Y_{1/4}Li_{1/4}TaO_3$, tetragonal for $Sm_{1/4}Li_{1/4}TaO_3$ and $Nd_{1/4}Li_{1/4}TaO_3$ and cubic for $La_{1/4}Li_{1/4}TaO_3$.
2. With decreasing the subcell volume, the lithium ion conductivity decreased and the activation energy increased. The subcell volume controls the bottleneck size for lithium ion transport and the subcell volume is one of the major factors in influencing lithium ion conduction in these perovskite-type conductors.
3. The perovskites with the ordering of cations and vacancies at the A-sites exhibited higher activation energy and lower mobility than the cubic perovskites with the same subcell volume.

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